

Final Related Literature and Studies

Related Literature

According to **Okaka (2022)** in the study *A Systematic Literature Review of Web-Based Student Response Systems: Advantages and Challenges*, web-based student response systems provide an effective approach for gathering real-time feedback from students. The review highlighted that these systems increase student participation, promote interactive learning, and allow instructors to receive immediate responses that can help improve teaching methods. Although the study identified challenges such as technological limitations and system usability, its findings support the development of the proposed **Web-based Centralized Faculty Evaluation System** by demonstrating that a web-based platform can effectively collect, process, and manage student evaluations while improving the efficiency of the evaluation process.

Similarly, the literature review *Online Peer Feedback in Higher Education: A Synthesis of the Literature* by **Taghizadeh Kerman, Banihashem, Karami, Er, van Ginkel, and Noroozi (2023)** emphasized the importance of online feedback platforms in creating collaborative learning environments. The researchers explained that digital feedback systems encourage active student participation while enabling instructors and administrators to monitor and analyze responses efficiently. These findings are directly relevant to the proposed system because it provides students with an accessible online platform to evaluate faculty members and allows administrators to centralize and analyze evaluation results for decision-making and faculty development.

Moreover, **Babik, Gehringer, Kidd, Sunday, Tinapple, and Gilbert (2024)**, in *A Systematic Review of Educational Online Peer-Review and Assessment Systems: Charting the Landscape*, discussed how web-based assessment systems improve the speed of data collection, organization of records, and analysis of feedback compared with traditional paper-based methods. These findings support the proposed **Web-based Centralized Faculty Evaluation System**, which replaces manual evaluation procedures with an online centralized database, reducing paperwork, minimizing data redundancy, and providing faster generation of evaluation reports.

Another systematic review published in **2025**, *Students' Perceptions and Outcome of Teacher Feedback: A Systematic Review*, examined how teacher feedback influences student engagement and academic performance. The study found that structured feedback mechanisms strengthen communication between students and educators while helping institutions identify opportunities for academic improvement. This supports the proposed study because the system will provide a structured and organized process for collecting student evaluations, enabling school administrators to use the results to improve teaching effectiveness and overall educational quality.

Furthermore, **Trujillo et al. (2025)**, in *The Current Landscape of Formative Assessment and Feedback in Graduate Studies: A Systematic Literature Review*, emphasized that structured feedback systems enhance student learning, instructor performance, and continuous academic improvement. The researchers concluded that timely and organized feedback allows educators to monitor performance and make appropriate improvements. Likewise, the

proposed **Web-based Centralized Faculty Evaluation System** is designed to facilitate timely faculty evaluations, securely store evaluation records in a centralized database, and provide meaningful reports that support evidence-based decisions for faculty performance improvement.

Related Studies

The study **Web-Based Student Academic Performance Evaluation System Using Data Mining (2022)** developed a system that analyzes student data to evaluate academic performance trends. The system demonstrated that digital technologies can improve data accuracy and support informed decision-making through automated analysis. Although the study focused on student performance rather than faculty evaluation, it supports the proposed system by illustrating how centralized digital data can be analyzed efficiently to generate reliable reports that assist institutional decision-making.

The study **Enhancing UX/UI: A Mixed-Approach Evaluation of a Web-Based Student Clearance System (2024)** evaluated a web-based student clearance system implemented in a Philippine university. Using a mixed-methods approach, the researchers found that the system improved usability, efficiency, and user satisfaction while simplifying administrative processes. These findings are relevant to the proposed **Web-based Centralized Faculty Evaluation System** because both systems aim to replace manual procedures with an accessible and user-friendly web application that enhances institutional efficiency and improves the user experience.

The study **Web-Based Student Learning Activities and Development Information System Using PIECES Analysis (2025)** developed a web-based platform for monitoring student learning activities and academic development. The results showed that the system improved information management, monitoring efficiency, and record organization. Similarly, the proposed faculty evaluation system applies centralized data management principles to organize evaluation records, simplify report generation, and improve the efficiency of evaluation administration.

In **A Web-Based Student Learning Outcome Assessment System for Secondary Education: Design, Development, and Empirical Evaluation (2025)** by **Sadhana and Freire**, the researchers designed and evaluated a web-based assessment platform that demonstrated high levels of usability, reliability, and effectiveness. The study supports the proposed research because it confirms that web-based assessment systems can successfully automate educational evaluation processes while improving the accuracy, accessibility, and management of assessment data. These principles are likewise applied in the proposed faculty evaluation system.

Finally, the study **Intelligent Web-Based Student Performance Prediction Using Machine Learning in Java (2025)** introduced a web application that predicts student performance using machine learning techniques. The research demonstrated how centralized digital records and automated analysis provide valuable insights for educational planning. While the proposed study does not implement machine learning, it similarly utilizes centralized digital data to produce accurate faculty evaluation reports, helping

administrators make informed decisions regarding teaching effectiveness and faculty development.

Conceptual Framework (Input–Process–Output Model)

The study adopts the Input–Process–Output (IPO) Model as its conceptual framework. The model illustrates how the required inputs are transformed through a series of development processes to produce the proposed Web-based Centralized Faculty Evaluation System.

Input

The inputs consist of the information, requirements, and resources needed in developing the proposed system. These include:

- Existing faculty evaluation process
- Requirements gathered from the School Head/Education Office (EO)
- Review of related literature and studies
- School policies and faculty evaluation criteria
- Software and hardware requirements
- Development tools and technologies

Process

The process includes the activities performed during the development of the proposed system. These include:

- Planning
- Requirements gathering
- System analysis
- System design
- System development
- System testing
- Implementation
- Evaluation and revision

Output

The output is the completed Web-based Centralized Faculty Evaluation System, which provides:

- Online faculty evaluation
- Centralized evaluation database
- Automated report generation
- Secure user authentication
- Efficient data management
- Faster retrieval of evaluation records
- Improved decision-making for administrators

Final Review of Research Methodology

The review of the research methodology is important to ensure that the development process of the proposed Web-based Centralized Faculty Evaluation System is systematic and aligned with the objectives of the study. The researchers employed a structured software development methodology to analyze, design, develop, test, and evaluate the proposed system. This methodology ensures that the system meets the requirements of the institution while providing an efficient and reliable faculty evaluation platform.

Methodology Stages Involved

Requirements Analysis

The researchers analyzed the existing faculty evaluation process through consultation with the **School Head through the Education Office (EO)**. The gathered requirements became the basis for identifying the features and functionalities of the proposed system.

System Design

The researchers designed the system architecture, database, user interface, and system workflow. The design focused on creating a secure, organized, and user-friendly faculty evaluation system.

System Development

The proposed **Web-based Centralized Faculty Evaluation System** was developed using modern web technologies. The front-end was built using **React, HTML, TypeScript, and Tailwind CSS**, while the back-end was developed using **Node.js, Express.js, REST API, and MySQL**.

Testing and Evaluation

The developed system was tested to ensure its functionality, usability, reliability, and efficiency. Testing covered the following system modules:

- User Authentication
- Student Dashboard
- Faculty Evaluation
- Evaluation Form Management
- Faculty Management
- User Management
- Report Generation
- Centralized Data Management

Feasibility Analysis

The feasibility of the proposed system was evaluated to determine whether it could be successfully implemented within the institution.

- **Technical Feasibility** – Evaluates whether the available hardware, software, and development tools are sufficient to develop and operate the proposed system.
- **Economic Feasibility** – Determines whether the proposed system is cost-effective by utilizing existing resources and minimizing operational expenses.
- **Operational Feasibility** – Assesses whether students and administrators can effectively use the system to perform faculty evaluations.
- **Legal Feasibility** – Ensures that the proposed system complies with applicable laws and regulations, including the **Data Privacy Act of 2012 (Republic Act No. 10173)**.

- **Schedule Feasibility** – Determines whether the project can be completed within the planned development timeline.